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March 2026

STATISTICS AND BIG DATA II

GROUP PROJECT

AIRBUS

A) Data description and justification of selected variables

1. Dependant variable and Sample design:

For this project, our team chose Topic 1" Stock Market Prediction". Specifically, we aim to predict the weekly returns of Airbus SE, one of the world's largest aerospace and defense companies.

The dependent variable is the weekly simple return of Airbus, computed as $r_t = (P_t - P_{t-1}) / P_{t-1}$. Simple returns offer a simple interpretation as percentage price changes, making them intuitive for comparing forecasting performance across models. The dataset spans from January 2016 to January 2026 at weekly frequency, providing 522 observations that we selected from Bloomberg. This ten-year window captures multiple market regimes; the pre-COVID expansion, the 2020 pandemic shock, the post-pandemic recovery, the 2022 inflationary environment, and more recent monetary tightening cycles. All variables are aligned to the same weekly frequency to ensure temporal homogeneity.

We selected 8 exogenous variables, each capturing a distinct economic channel relevant to *Airbus* returns. *Boeing* is included as Airbus's direct competitor in the global aviation duopoly, capturing competitive dynamics and sector-specific shocks. *Safran* is a key supplier through its CFM International joint venture and reflects supply chain conditions directly tied to Airbus production. The *CAC 40 Index* controls for systematic risk in European equities, as Airbus is a major constituent exposed to broader market sentiment. The *EUR/USD* exchange rate is a critical profitability driver since Airbus incurs costs in euros but sells aircraft in US dollars. The *VIX Index* serves as a proxy for global risk aversion, to which Airbus is particularly sensitive as a cyclical stock with long-duration cash flows. *Brent Crude Oil* operates through a dual channel: higher oil prices squeeze airline profitability but also incentivize investment in fuel-efficient aircraft, benefiting Airbus's newer fleet. The *US 10-Year Treasury Yield* captures monetary policy expectations and discount rate effects on capital-intensive firms with long production cycles. Finally, *General Electric*, through GE Aerospace, complements Safran as a gauge of aerospace supply chain health.

2. Descriptive Statistics

	Mean	Median	Std.Dev	Min	Max	Skewness	Kurtosis
AIR	0.0034	0.0040	0.0490	-0.2738	0.4282	0.7434	18.4858
BA	0.0029	0.0018	0.0665	-0.4418	0.7051	1.7327	33.8479
SAF	0.0040	0.0059	0.0448	-0.2759	0.3142	-0.1654	15.6721
CAC	0.0014	0.0023	0.0256	-0.1986	0.1070	-1.1089	11.7541
EURUSD	0.0002	-0.0002	0.0101	-0.0377	0.0441	0.3458	4.9619
VIX	0.0127	-0.0113	0.1734	-0.4266	1.3484	1.9725	12.9787
BRENT	0.0024	0.0045	0.0535	-0.2523	0.3682	0.3396	9.7765
US10YR	0.0029	-0.0004	0.0588	-0.3363	0.3716	0.6531	10.2956
GE	0.0026	0.0024	0.0494	-0.1779	0.1994	-0.0789	5.0431

Table 1 reports the descriptive statistics of weekly returns over the sample period. All mean returns are positive and close to zero, typical for weekly financial data. Airbus exhibits a mean weekly return of 0.34%, slightly below Safran (0.40%) but above Boeing (0.29%). In terms of volatility, Boeing is the most volatile equity (6.65%), followed by Airbus (4.90%) and Safran (4.48%), while the CAC 40 exhibits the lowest equity volatility (2.56%), reflecting diversification. The skewness and kurtosis figures reveal significant departures from normality. Boeing shows extreme kurtosis (33.85) driven by COVID era outliers, and all variables display kurtosis well above 3, confirming leptokurtic distributions with fat tails.

3. Figures: Visual Analysis and Correlation Structure

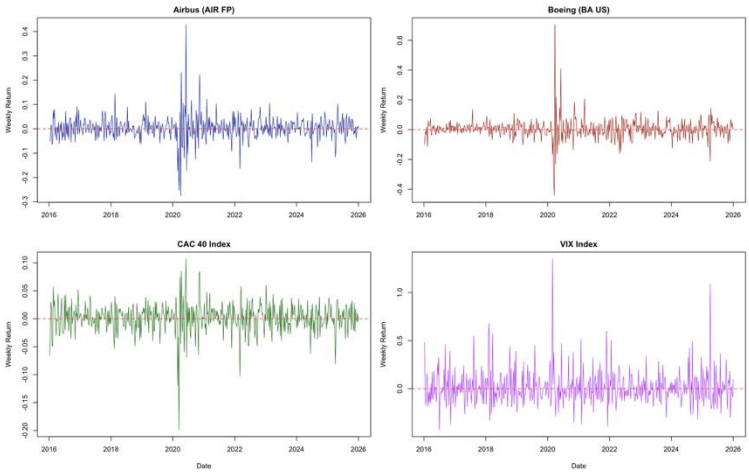


Figure 1 displays the weekly returns of Airbus, Boeing, CAC 40, and the VIX over the 2016-2026 period. All series exhibit volatility clustering, with the most striking episode around early 2020 during the COVID-19 shock. During this period, Airbus and Boeing experienced extreme negative returns of approximately -30% and -40% in a single week respectively, while the VIX spiked above +100% reflecting a sudden surge in market fear. The CAC 40 also recorded its sharpest weekly decline at around -20%. The VIX panel confirms the inverse relationship between risk appetite and equity return, its largest positive spikes coincide precisely with the deepest drawdowns in the three other panels. Outside this episode, returns are generally centered around zero, suggesting stationary behavior suitable for econometric modeling.

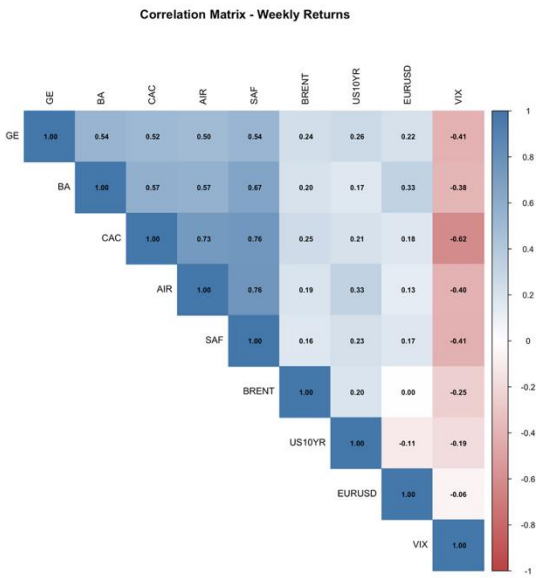


Figure 2 presents the correlation matrix of weekly returns across all variables over the 2016-2026 sample period. The strongest positive correlations with Airbus are found with Safran (0.76) and the CAC 40 (0.73), reflecting supply chain linkages and systematic risk exposure respectively. Boeing also exhibits a substantial positive correlation with Airbus (0.57), confirming that sector-wide shocks tend to affect both competitors simultaneously despite their rivalry. The VIX displays the most pronounced negative correlation with Airbus (-0.40), consistent with its role as a fear gauge, suggesting that European equities are particularly sensitive to global risk sentiment. More broadly, the VIX is negatively correlated with all equity variables Among the remaining variables, the US 10-Year yield shows a moderate positive correlation with Airbus (0.33), indicating that rising yields tend to coincide with positive Airbus returns. Importantly, while some variables are correlated with each other no pair of exogenous variables exceeds a correlation of 0.80, which limits multicollinearity concerns in our regression models. This will nevertheless be monitored through Variance Inflation Factors (VIF) in the modeling stage.

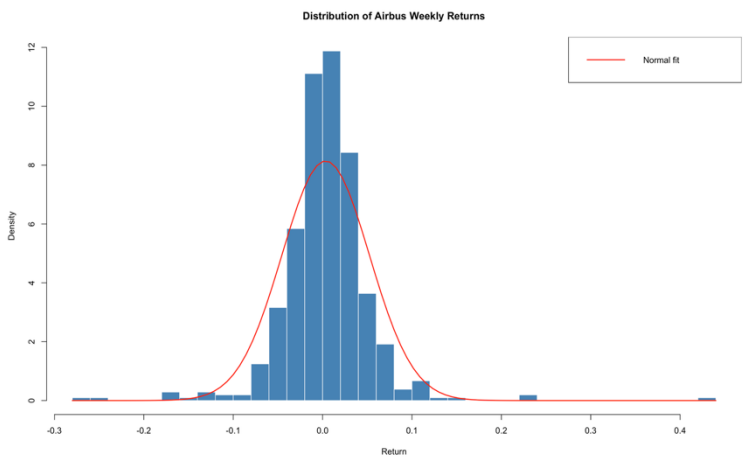


Figure 3 displays the distribution of Airbus weekly returns alongside a fitted normal curve. The histogram confirms the findings from Table 1, the distribution is centered slightly above zero, consistent with a positive mean return, but exhibits a much sharper peak and heavier tails than the normal distribution. Several extreme observations are visible beyond 20%, corresponding to the COVID-19 shock period. This departure from normality, characterized by excess kurtosis and positive skewness, suggests that standard linear models assuming Gaussian errors may underperform, and supports the exploration of more flexible approaches in the modeling stage.

B) Modeling and forecasting experiment

1. **Train and Test samples:**

To evaluate out-of-sample forecasting performance, we split the dataset chronologically, the training set covers January 2016 to December 2023 (approximately 80% of observations), while the test set spans January 2024 to January 2026 (approximately 20%). A chronological split, rather than a random one, is essential for time series data to avoid look-ahead bias. The training set is used for parameter estimation, while the test set is reserved exclusively for evaluating predictive accuracy on unseen data.

2. **Choice of Models and Evaluation Criteria:**

We estimated six models of increasing complexity, spanning three families of methods: linear models, regularization, and tree-based methods. This progression allows us to assess whether added flexibility and non-linearity translate into meaningful gains in predictive power.

LINEAR MODELS:

Multiple Linear Regression **(OLS)** serves as our baseline. It assumes a linear relationship between Airbus returns and the 8 exogenous variables. The objective is to establish a benchmark against which more complex models are compared, and to identify which predictors are statistically significant via t-tests and p-values.

Lasso Regression extends OLS by adding an L1 penalty that shrinks some coefficients to exactly zero, performing automatic variable selection. The objective is to determine whether a sparser model using fewer predictors can match or outperform the full OLS specification. The optimal lambda (0.00188) was selected via 10-fold cross-validation on the training set.

NON-LINEAR MODELS :

The Generalized Additive Model **(GAM)** replaces each linear term with a smooth non-parametric function fitted via smoothing splines with 4 degrees of freedom per predictor. The

objective is to test whether non-linear relationships exist between the predictors and Airbus returns, which the OLS would fail to capture.

ENSEMBLE TREES :

Bagging aggregates 500 regression trees, each trained on a bootstrap sample using all 8 predictors at each split (mtry = p). By averaging across many trees, it reduces variance compared to a single tree. The objective is to serve as a bridge between individual decision trees and the Random Forest.

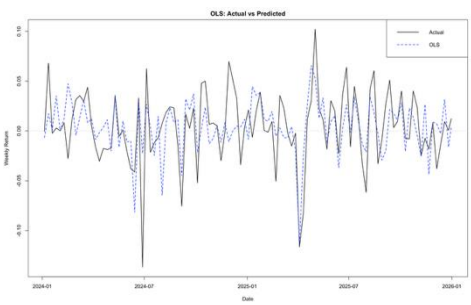
Random Forest aggregates 500 decorrelated regression trees, each trained on a bootstrap sample using a random subset of 3 predictors at each split (mtry = $\sqrt{p}$, the standard rule for regression). The objective is to reduce variance through ensemble averaging and to provide a variable importance ranking identifying the key drivers of Airbus returns.

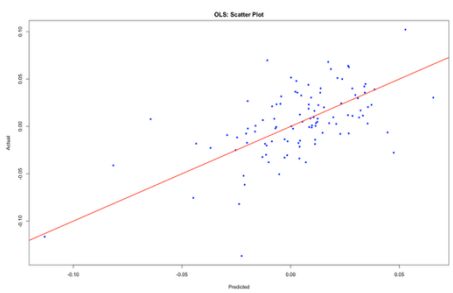
Gradient Boosting **(GBM)** builds trees sequentially with an interaction depth of 4 and a shrinkage rate of 0.01, where each new tree corrects the errors of the previous ensemble. The optimal number of trees (1,748) was determined via 5-fold cross-validation. The objective is to test whether this iterative error-correction mechanism further improves predictive accuracy beyond the Random Forest.

Model performance is assessed using five metrics; MSE (Mean Squared Error), which penalizes large forecast errors quadratically; RMSE (Root MSE), expressed in the same unit as returns for easier interpretation; MAE (Mean Absolute Error), which gives equal weight to all errors regardless of magnitude; out-of-sample R^2, which measures the share of return variance explained by the model; and Adjusted R^2, which corrects for the number of predictors by penalizing model complexity. We also compare in-sample R^2 (computed on the training set) with out-of-sample R^2 (computed on the test set) to detect potential overfitting: a large gap between the two signals that the model memorizes noise in the training data rather than learning generalizable patterns.

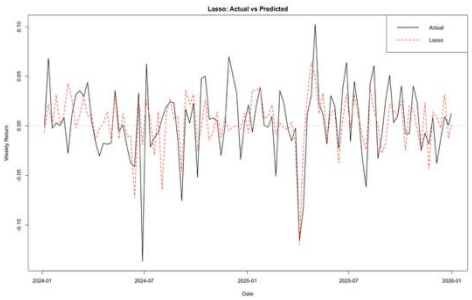
3. Results and comments

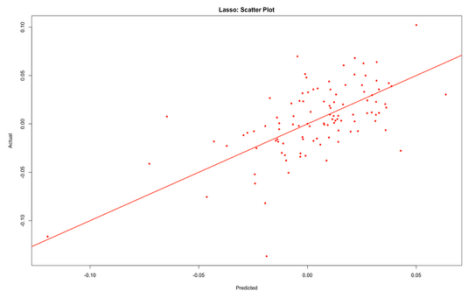
a) **Model 1: OLS.** The baseline achieves an out-of-sample R^2 of 0.324 (RMSE = 0.0298, MAE = 0.0230). The in-sample R^2 of 0.708 indicates a solid fit, with a moderate gap suggesting some generalization loss. The OLS summary reveals that Safran ($p < 0.001$), CAC 40 ($p < 0.001$), and US10YR ($p < 0.001$) are highly significant, while EUR/USD ($p = 0.41$) and BRENT ($p = 0.79$) are not. The Actual vs Predicted plot shows that OLS captures the general direction of returns but consistently underestimates extreme movements. The scatter plot confirms this: points are clustered around the 45-degree line for moderate returns but deviate substantially for larger values. These limitations motivate the use of regularized methods.



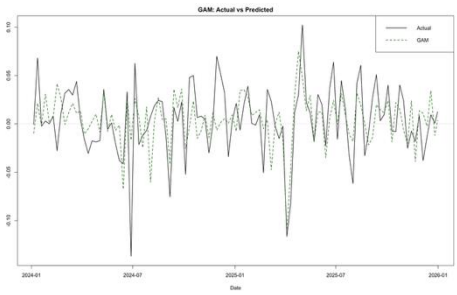


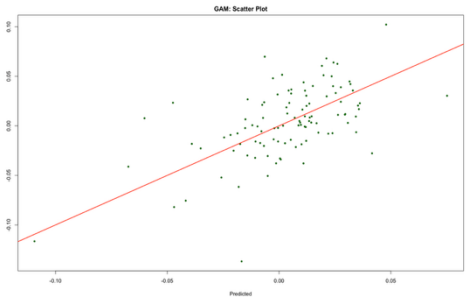
b) **Model 2: Lasso.** Building on OLS, the Lasso improves slightly on all metrics (RMSE = 0.0292, R^2 = 0.352) while using a simpler model. Its in-sample R^2 (0.701) is comparable to OLS, confirming that regularization does not sacrifice fit. The key finding is that the Lasso eliminates three variables: EUR/USD, VIX, and BRENT, retaining only SAF (0.45), CAC (0.61), US10YR (0.11), BA (0.04), and GE (0.03). This confirms the insignificance of these variables already visible in the OLS p-values. The cross-validation plot shows that the MSE stabilizes quickly as lambda decreases, indicating that only a small amount of regularization is needed. The scatter plot shows a tighter clustering around the diagonal compared to OLS, consistent with the improved R^2. However, the Lasso remains linear to test whether non-linear relationships could further improve predictions, we turn to the GAM.



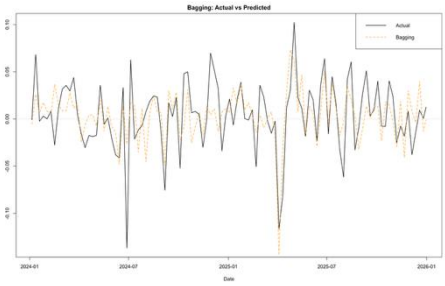


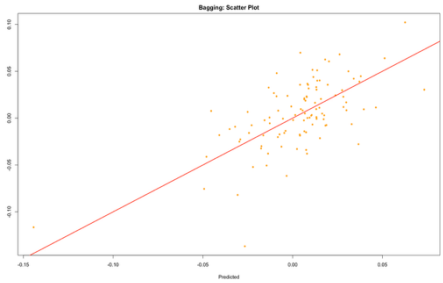
c) **Model 3: GAM.** Despite its added flexibility, the GAM does not improve upon OLS (RMSE = 0.0299, R^2 = 0.321) and has the highest in-sample R^2 among linear-type models (0.781). This gap of 0.46 between in-sample and out-of-sample suggests that the smoothing splines overfit the training data. The partial effect plots confirm this interpretation: the smooth functions for Safran and CAC 40 are nearly linear, while weaker predictors (VIX, BRENT, EUR/USD) show wide confidence bands indicating poor signal. The scatter plot displays more dispersion than OLS and Lasso. This result indicates that non-linearity at the individual predictor level is not a major source of improvement, motivating the shift to tree-based ensemble methods that capture interactions between variables.



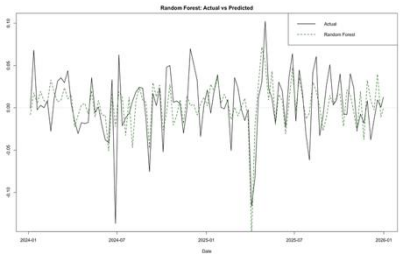


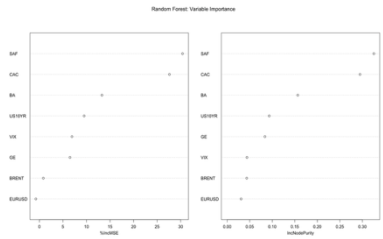
d) **Model 4: Bagging.** The transition to ensemble methods yields a significant improvement. Bagging achieves the best performance across all metrics (RMSE = 0.0282, MAE = 0.0222, R^2 = 0.397), outperforming all linear approaches. Its in-sample R^2 of 0.929 is very high, yet this translates into the best out-of-sample performance, demonstrating that averaging 500 trees effectively reduces variance without overfitting. The Actual vs Predicted plot shows that Bagging tracks the actual returns most closely among all models. The scatter plot confirms the tightest clustering around the 45-degree line. With 8 predictors, using all variables at each split (mtry = 8) proves more effective than the random subset approach, as no variable is arbitrarily excluded from important splits.

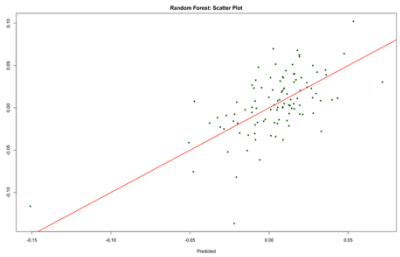




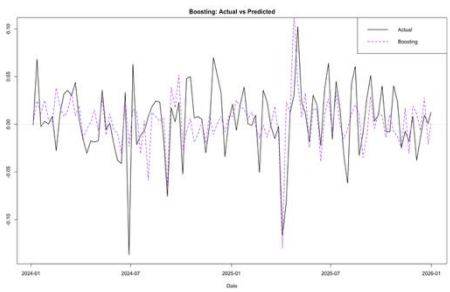
e) **Model 5: Random Forest.** The Random Forest performs well (RMSE = 0.0287, R^2 = 0.376) but slightly below Bagging on all metrics. The difference lies in the mtry parameter: by using only 3 predictors at each split instead of 8, the Random Forest decorrelates its trees but occasionally misses key variables like Safran or CAC 40 at critical splits. Its in-sample R^2 (0.929) is virtually identical to Bagging, confirming that both models capture similar training patterns, the difference emerges only out-of-sample. The variable importance ranking identifies Safran and CAC 40 as the dominant predictors, followed by Boeing and US10YR, while EUR/USD and VIX contribute the least, fully consistent with the Lasso's variable selection.

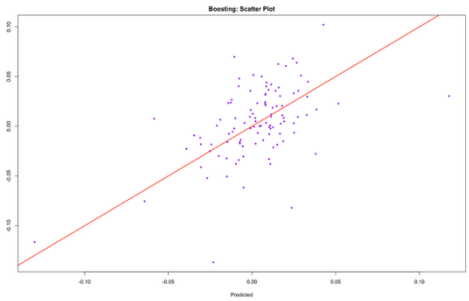






f) **Model 6: Boosting.** Surprisingly, Boosting performs worst among all models (RMSE = 0.0320, R^2 = 0.221). The in-sample R^2 of 0.874 is high but the out-of-sample R^2 is the lowest, a gap of 0.65, the largest of any model. This is a textbook case of overfitting: the sequential error-correction mechanism with 1,748 trees memorizes noise rather than learning generalizable patterns. The Actual vs Predicted plot shows the most erratic predictions, and the scatter plot confirms the widest dispersion around the diagonal. Despite having the most sophisticated architecture, Boosting is counterproductive on noisy weekly return data with only 8 predictors.

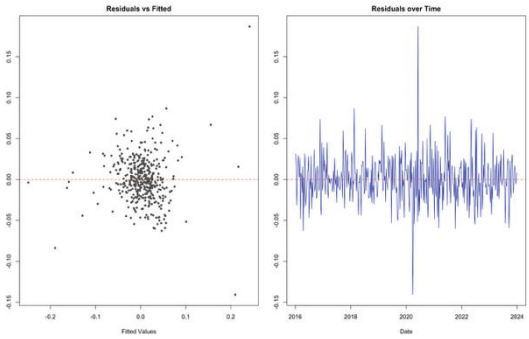




4. Robustness check

We conducted two types of robustness checks: diagnostic tests on the OLS residuals to verify underlying assumptions, and an alternative train-test split to confirm model ranking stability.

a) **Statistical diagnostics on OLS.** Four tests were performed on the OLS residuals. The Shapiro-Wilk test ($p < 0.001$) and the Jarque-Bera test ($p < 0.001$) both reject normality, consistent with the fat tails documented in Part 1. The Ljung-Box test ($p = 0.008$) detects significant autocorrelation in the residuals, meaning that OLS prediction errors from one week contain information about the next. The Breusch-Pagan test ($p < 0.001$) confirms heteroscedasticity: the variance of residuals is not constant, being larger during volatile periods. These three violations justify the use of methods that do not rely on these assumptions, the Lasso, Bagging, Random Forest, and Boosting are all robust to such violations by construction.



b) **Alternative split (70/30).** To verify that our model ranking is not driven by the train-test boundary, we re-estimate all six models using a 70/30 split (training through December 2022, testing from January 2023). Tables 3 and 5 compare the results across both configurations. The Bagging model remains the best-performing model under both splits, with the lowest MSE, RMSE, and MAE, and the highest R^2 (0.410 under 70/30 vs 0.397 under 80/20). The overall ordering: Bagging, Random Forest, Lasso, OLS, GAM, Boosting is preserved, confirming that our findings are robust to the choice of train-test boundary.

Table 3: Robustness Check - 80/20 vs 70/30 Split

Model	MSE_80_20	MSE_70_30	RMSE_80_20	RMSE_70_30	MAE_80_20	MAE_70_30
OLS	0.000890	0.000761	0.029834	0.027589	0.022952	0.021097
Lasso	0.000853	0.000738	0.029204	0.027172	0.022479	0.020757
GAM	0.000894	0.000767	0.029893	0.027696	0.023123	0.021001
Bagging	**0.000793**	**0.000682**	**0.028169**	**0.026116**	**0.022153**	**0.020092**
Random Forest	0.000821	0.000714	0.028660	0.026721	0.022417	0.020329
Boosting	0.001025	0.000883	0.032018	0.029723	0.024025	0.021527

Table 5: Robustness Check - R^2 and Adjusted R^2 (80/20 vs 70/30)

Model	R2_80_20	R2_70_30	AdjR2_80_20	AdjR2_70_30
OLS	0.3238	0.3416	0.2675	0.3060
Lasso	0.3520	0.3614	0.2980	0.3268
GAM	0.3211	0.3365	0.2645	0.3006
Bagging	**0.3972**	**0.4100**	**0.3469**	**0.3781**
Random Forest	0.3760	0.3823	0.3240	0.3490
Boosting	0.2211	0.2358	0.1562	0.1945

C) Conclusion

Tables 2 and 4 summarize the out-of-sample forecasting performance and the in-sample vs out-of-sample R^2 & Ajd R^2 for all six models.

Table 2: Out-of-Sample Forecasting Performance (Test Set)

Model	MSE	RMSE	MAE
OLS	0.000890	0.029834	0.022952
Lasso	0.000853	0.029204	0.022479
GAM	0.000894	0.029893	0.023123
Bagging	**0.000793**	**0.028169**	**0.022153**
Random Forest	0.000821	0.028660	0.022417
Boosting	0.001025	0.032018	0.024025

Table 4: In-Sample vs Out-of-Sample R^2 and Adjusted R^2

Model	R2_InSample	AdjR2_InSample	R2_OutOfSample	AdjR2_OutOfSample
OLS	0.7076	0.7018	0.3238	0.2675
Lasso	0.7009	0.6950	0.3520	0.2980
GAM	0.7811	0.7768	0.3211	0.2645
Bagging	**0.9285**	**0.9271**	**0.3972**	**0.3469**
Random Forest	0.9286	0.9272	0.3760	0.3240
Boosting	0.8737	0.8713	0.2211	0.1562

Based on the results presented in Part B, we select the Bagging model as our best model for predicting Airbus weekly returns. It achieves the lowest forecasting errors (MSE = 0.000793, RMSE = 0.0282, MAE = 0.0222) and the highest out-of-sample R^2 (0.397) among all six models, under both the 80/20 and 70/30 splits.

Beyond objective metrics, this choice is supported by several subjective considerations. First, Bagging strikes an effective balance between complexity and performance. By using all predictors at each split, it avoids the information loss inherent in the Random Forest's random subset approach which proves counterproductive with only 8 variables. In contrast, the Boosting model, despite being the most complex approach, performed worst, illustrating that greater flexibility does not necessarily improve forecasting with noisy financial data.

Second, the consistency of findings across methods strengthens our confidence in the results. The Lasso's variable selection, the Random Forest's importance ranking, and the Boosting's relative influence all converge on the same conclusion; Airbus returns are primarily driven by Safran (supply chain) and the CAC 40 (systematic risk), while EUR/USD, VIX, and Brent contribute limited predictive power. This cross-model agreement suggests that these findings reflect genuine economic relationships rather than model-specific artifacts.

Finally, our analysis has several limitations. The weekly frequency may smooth out shorter-term dynamics, and the COVID-19 period represents an extreme structural break that could disproportionately influence estimation. Our feature set is limited to market-based variables and does not include firm-specific fundamentals such as order backlog or delivery data. Future research could explore time-varying models to account for regime changes or incorporate alternative data sources such as news sentiment to further improve predictive accuracy.